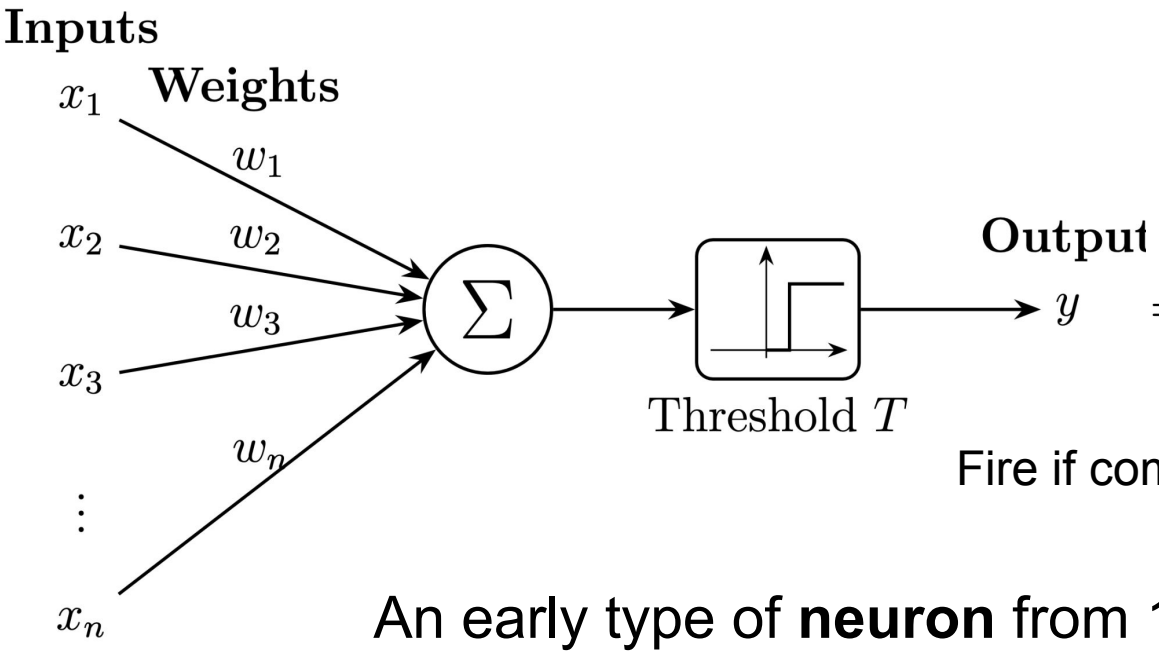
CS228: Introduction to Deep Learning

Lecture 3: perceptron

Instructor: Zhouxing Shi

Recap: Perceptron

Inputs combined linearly followed by a threshold logic



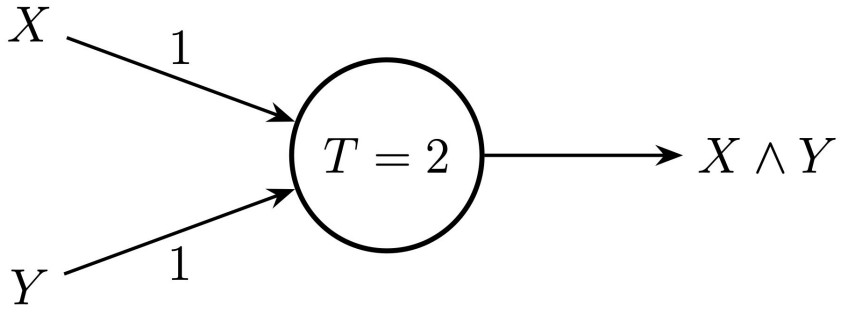
$$= \begin{cases} 1 & \text{if } \sum_i w_i x_i - T \geq 0 \\ 0 & \text{else} \end{cases}$$

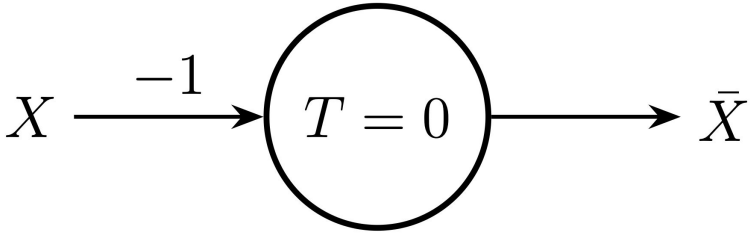
Fire if combined input exceeds threshold

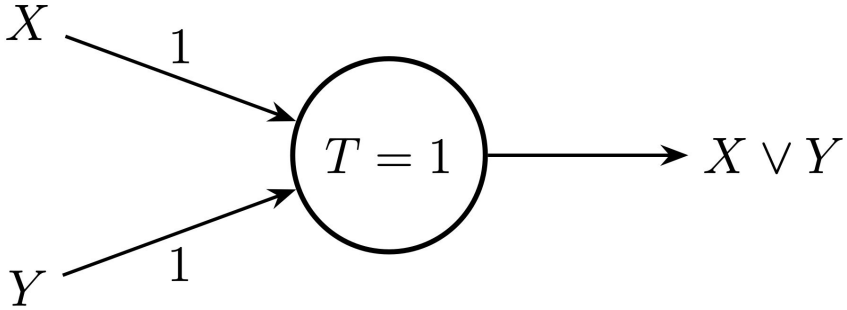
An early type of **neuron** from 1958

Recap: Boolean operations

A perceptron can implement boolean operations

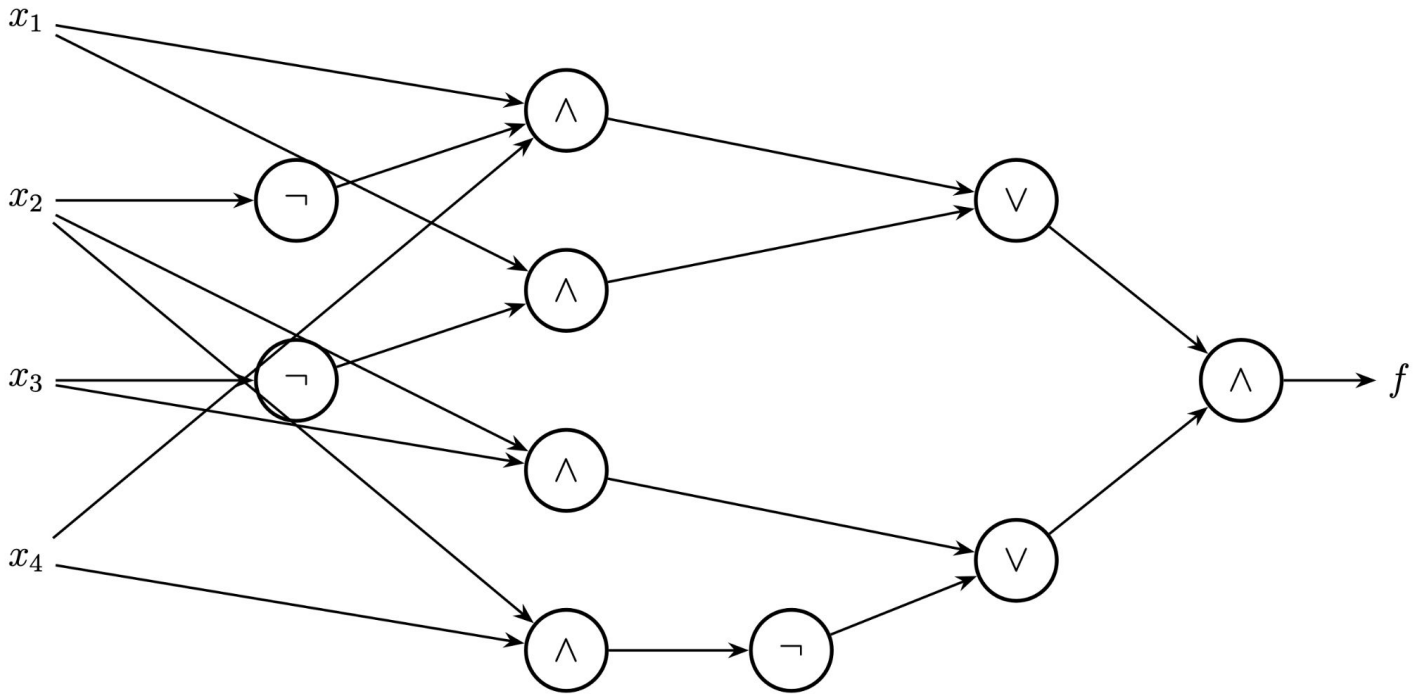






Recap: A more complex boolean function

$$f(x_1, x_2, x_3, x_4) = \Big((x_1 \,\&\, \bar{x}_2 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_3) \Big) \,\&\, \Big((x_2 \,\&\, x_3) \mid \overline{x_2 \,\&\, x_4} \Big)$$



Recap: A more complex boolean function

$$f(x_1, x_2, x_3, x_4) = \Big((x_1 \,\&\, \bar{x}_2 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_3) \Big) \,\&\, \Big((x_2 \,\&\, x_3) \mid \overline{x_2 \,\&\, x_4} \Big)$$

Truth table (showing rows outputting 1 only)

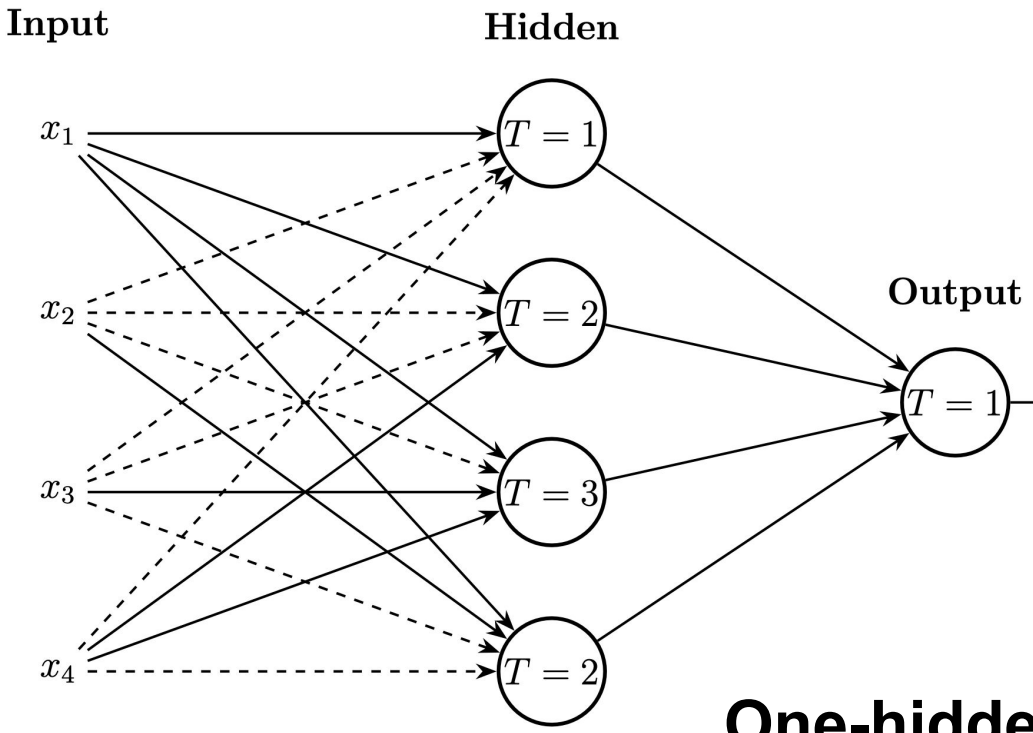
x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

$$\begin{aligned} f = {} & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \\ \mid {} & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \\ \mid {} & (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \\ \mid {} & (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \end{aligned}$$

Disjunctive Normal Form (DNF)

Any boolean function can be represented by a **truth table**

Recap: A more complex boolean function

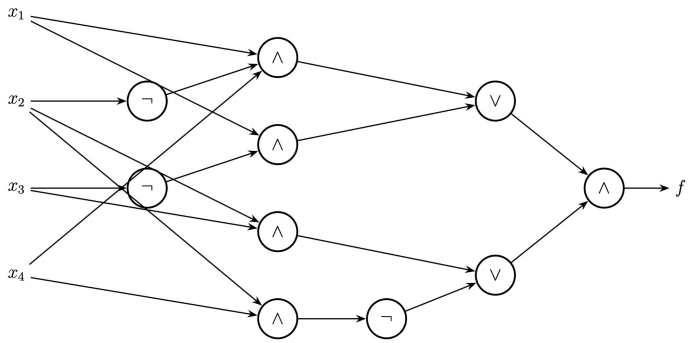


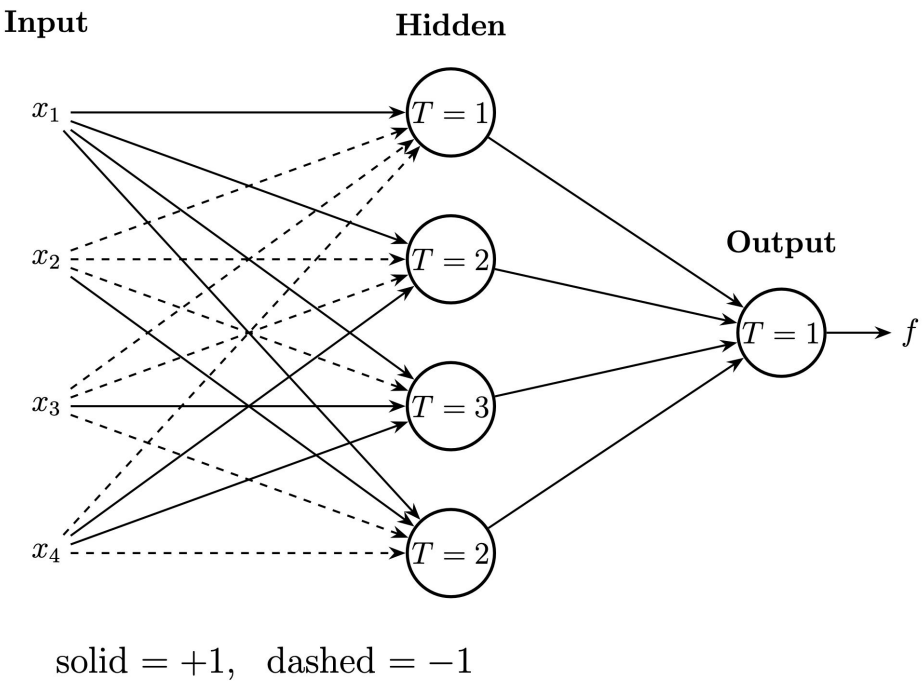
$$f = (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4)$$

solid $= +1$, dashed $= -1$

One-hidden-layer MLP is a Universal Boolean Function

Layered MLP

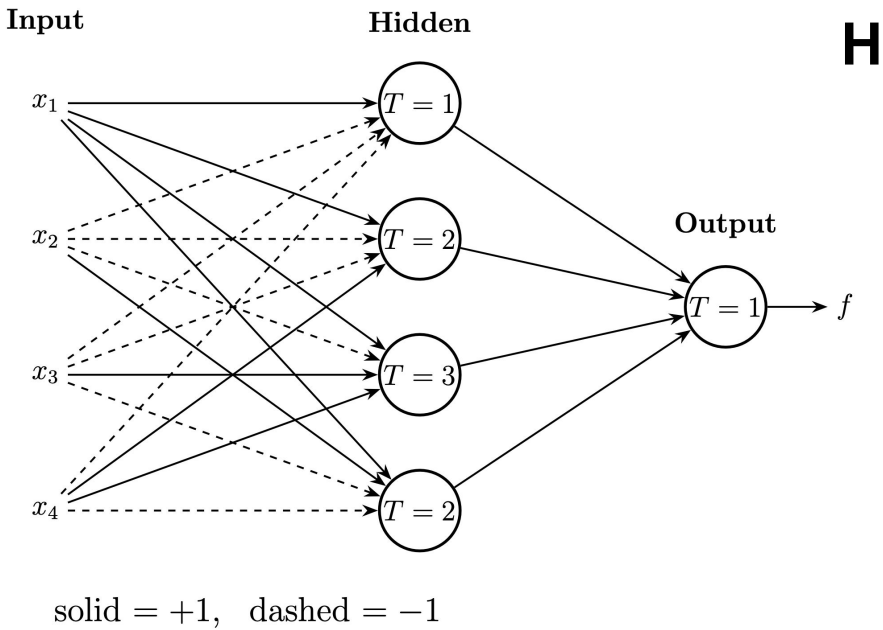




MLP with a single hidden layer

Universal Boolean Function

One-hidden-layer MLP for a boolean function with N inputs:



How many neurons do we need?

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

Universal Boolean Function

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

$$
\begin{aligned}
f = & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4)
\end{aligned}
$$

Karnaugh Map (K-map)

x_1x_2 \ x_3x_4	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	**1**	0	0	0
10	**1**	**1**	**1**	0

Universal Boolean Function

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

$$
\begin{aligned}
f = & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4)
\end{aligned}
$$

Karnaugh Map (K-map)

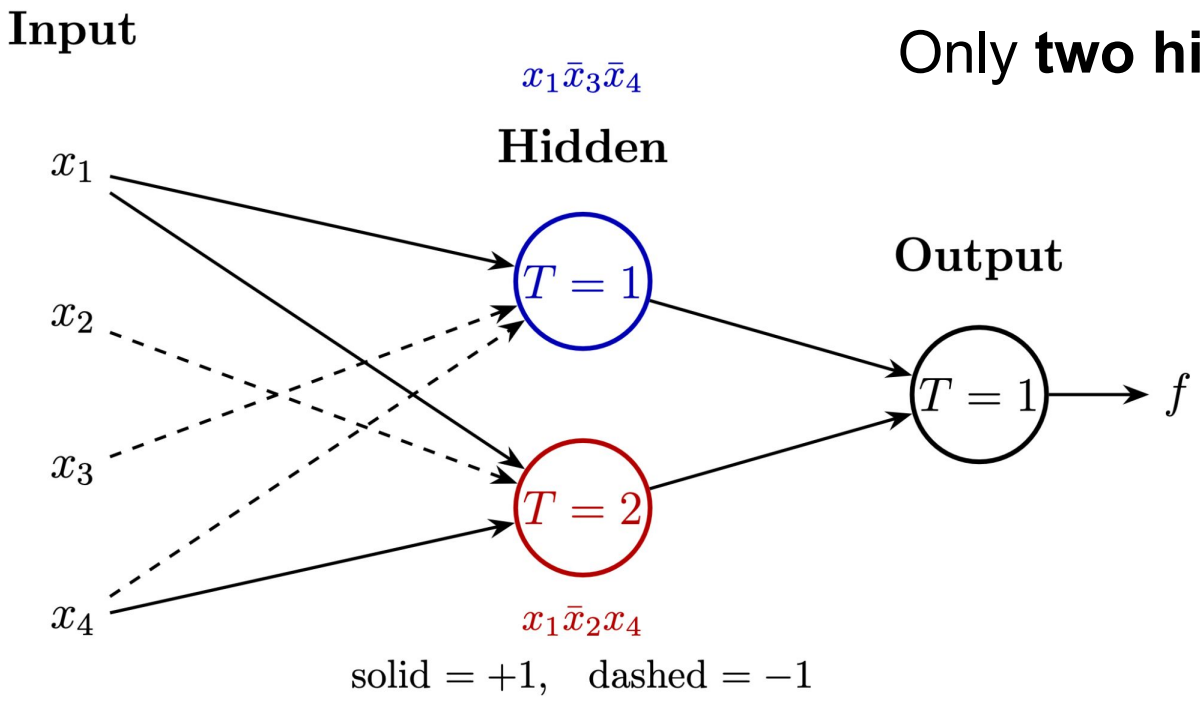
x_1x_2 \ x_3x_4	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	0	0	0
10	1	1	1	0

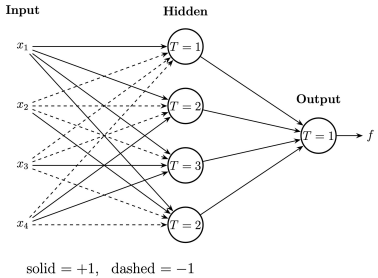
Reduced DNF $f = x_1\bar{x}_3\bar{x}_4 \mid x_1\bar{x}_2x_4$

Universal Boolean Function

$$f(x_1, x_2, x_3, x_4) = \Big((x_1 \,\&\, \bar{x}_2 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_3) \Big) \,\&\, \Big((x_2 \,\&\, x_3) \mid \overline{x_2 \,\&\, x_4} \Big)$$

Only **two hidden neurons** needed





Universal Boolean Function

One-hidden-layer MLP for a boolean function with N inputs:

Worst-case K-map

x_1x_2 \ x_3x_4	00	01	11	10
00	0	**1**	0	**1**
01	**1**	0	**1**	0
11	0	**1**	0	**1**
10	**1**	0	**1**	0

No adjacent 1s

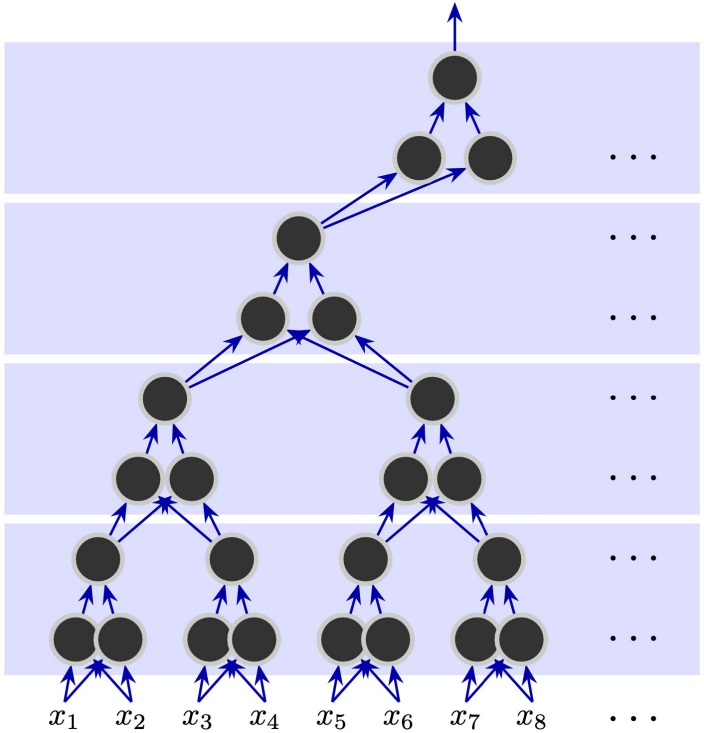
Need **up to 2^{N-1}** hidden neurons

Exponential in N

$$f = x_1 \oplus x_2 \oplus x_3 \oplus x_4$$

Universal Boolean Function

Deep MLP for a boolean function with N inputs:



$$f = x_1 \oplus x_2 \oplus \cdots \oplus x_N$$

Depth: $2\log_2 N$

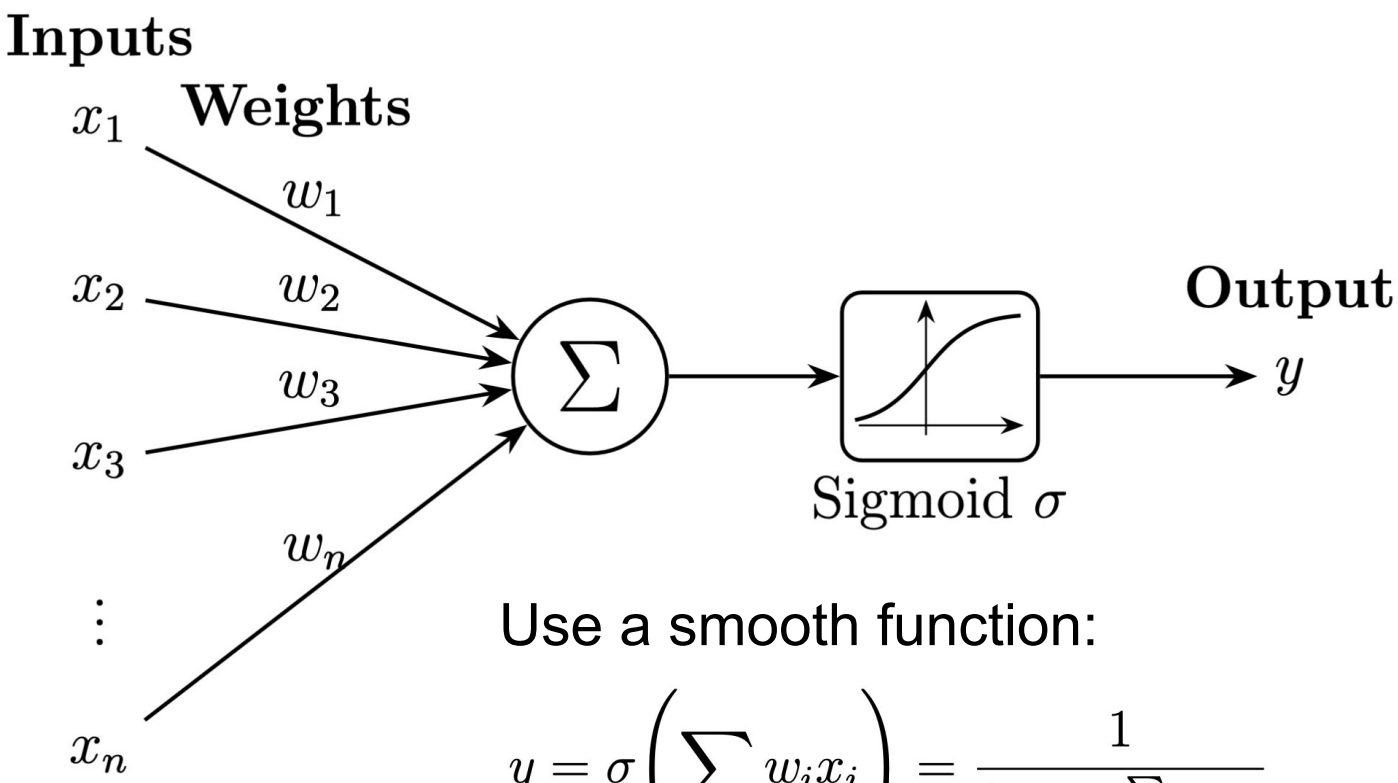
3(N-1) neurons

Size of deep MLP **scales linearly** with N for boolean functions

Universal Boolean Function

- MLPs are universal boolean functions
- A single-hidden-layer MLP is a universal boolean function
  - With sufficient width (could be exponentially large)
- Deep MLPs may require exponentially fewer neurons

Real value outputs

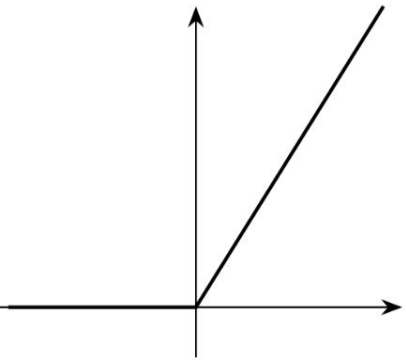


Use a smooth function:

$$y = \sigma\left(\sum_i w_i x_i\right) = \frac{1}{1 + e^{-\sum_i w_i x_i}}$$

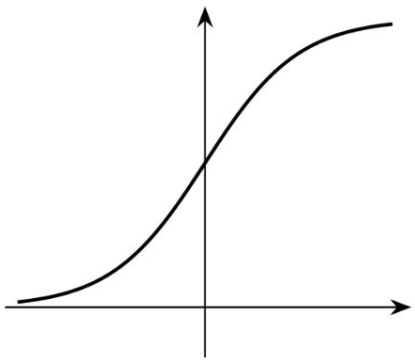
Activation functions

ReLU



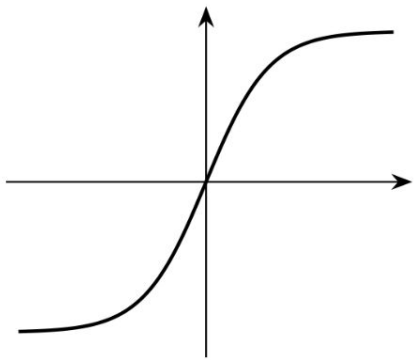
$$\max(0,\, x)$$

Sigmoid



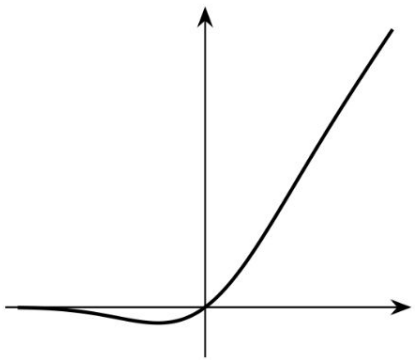
$$\frac{1}{1+e^{-x}}$$

Tanh



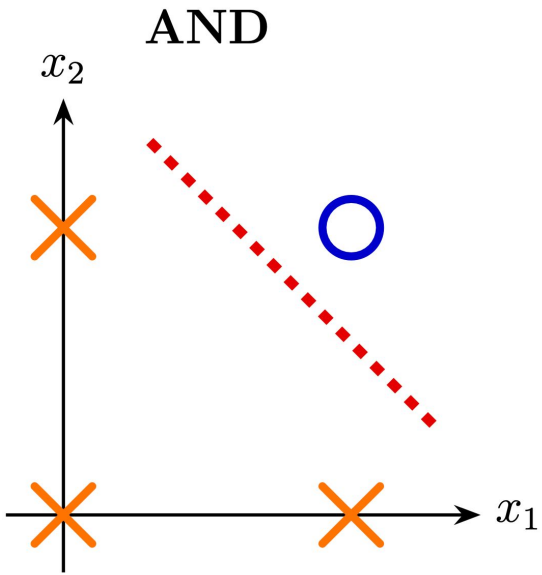
$$\tanh(x)$$

GeLU

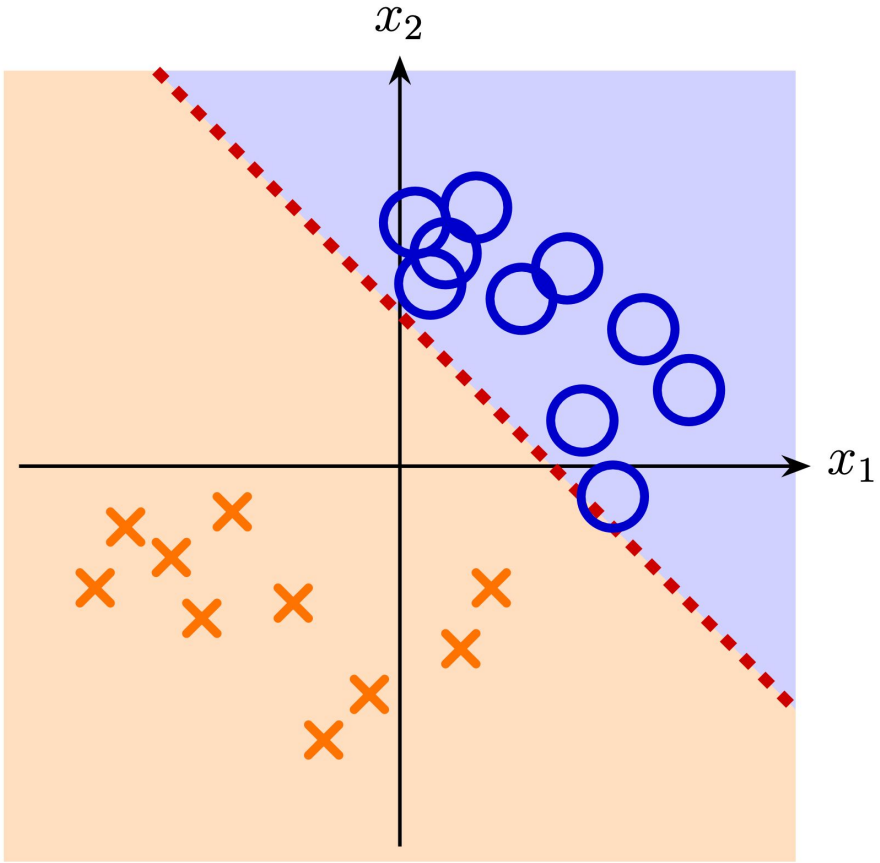


$$x \cdot \Phi(x)$$

MLP for classification



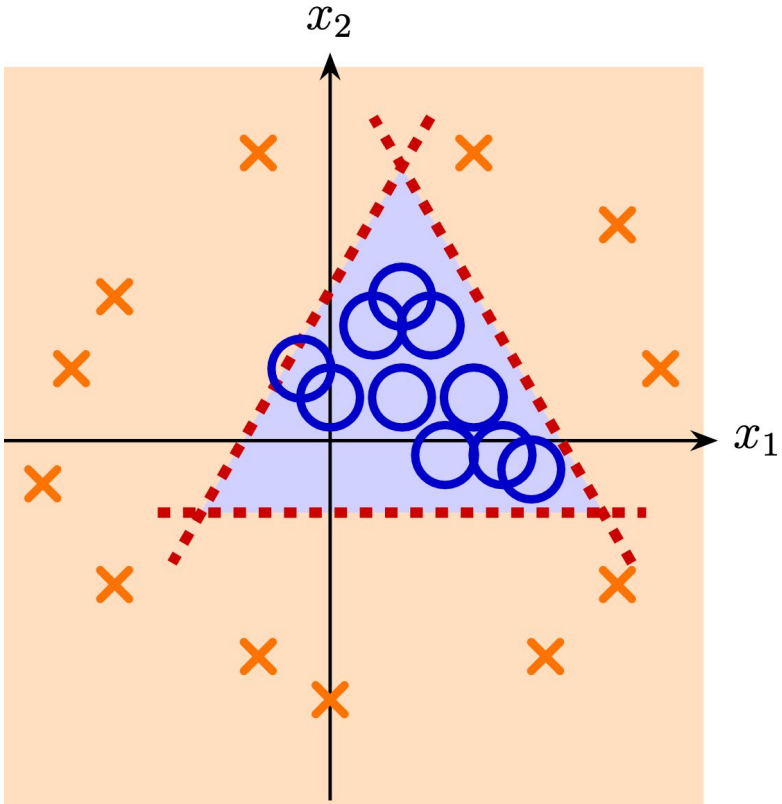
$x_1 \wedge x_2$



A single neuron needed

[Whiteboard]

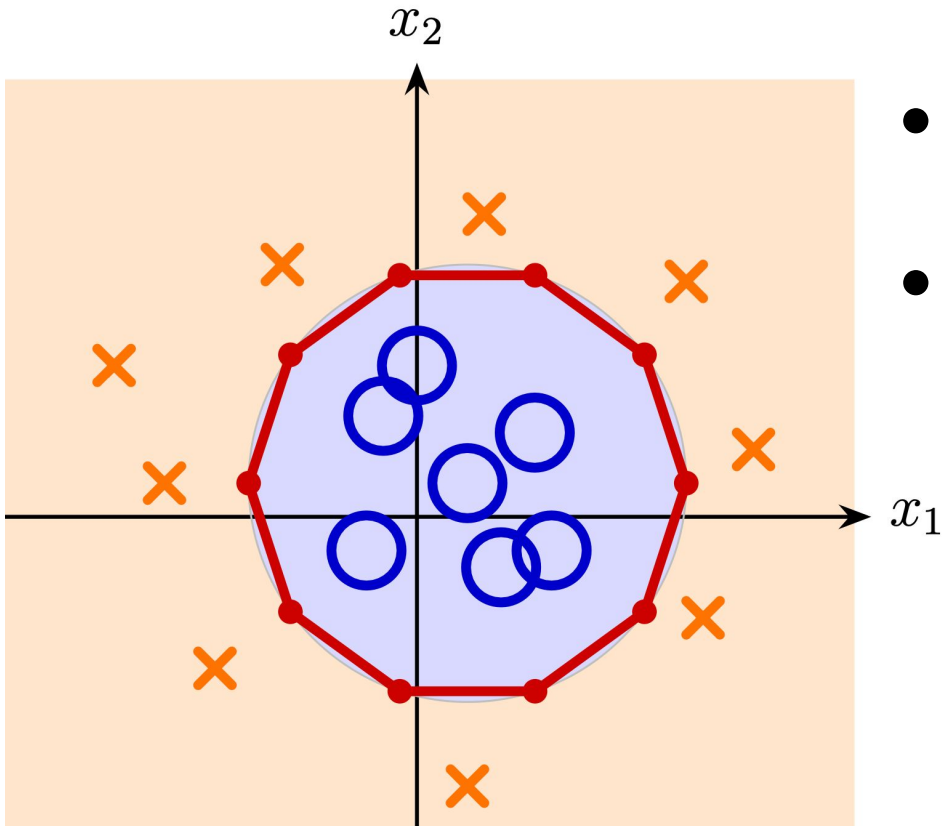
MLP for classification



Three hidden neurons

[Whiteboard]

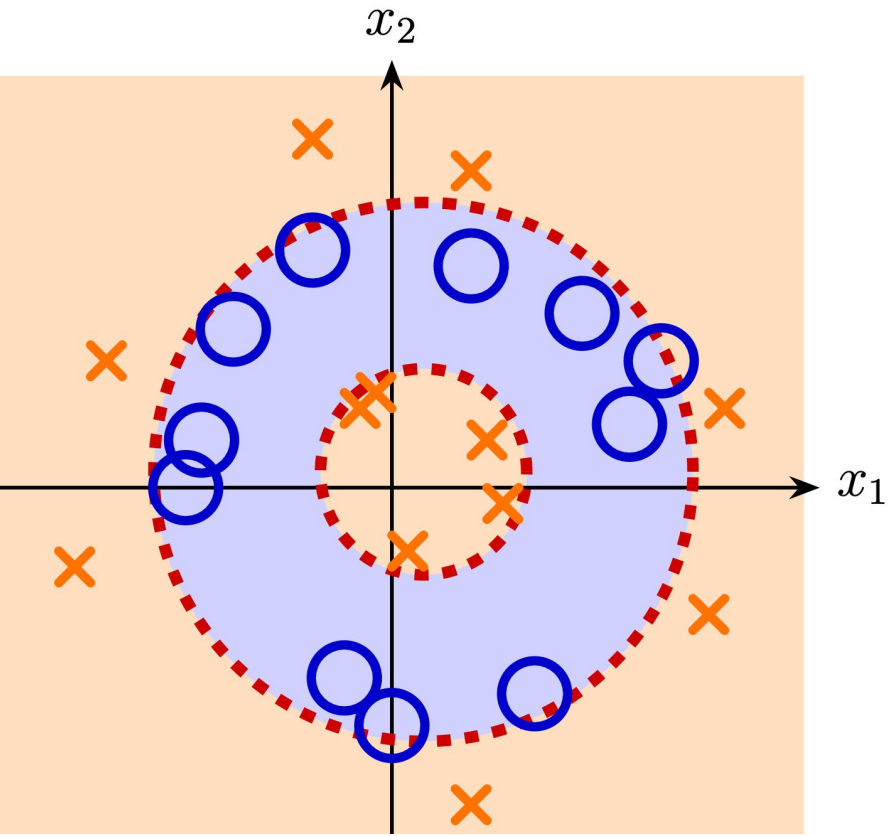
MLP for classification



- One hidden neuron for each edge
- A sufficiently large number of edges can approximate a circle

[Whiteboard]

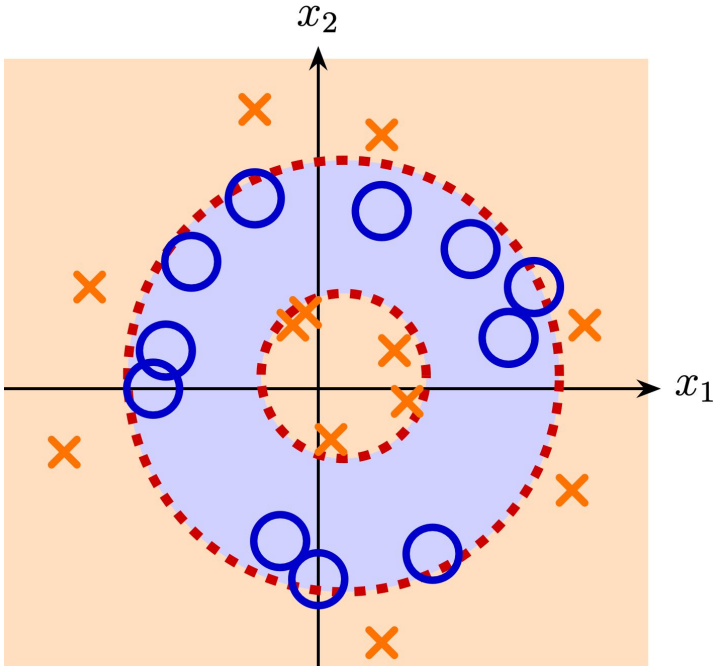
MLP for classification

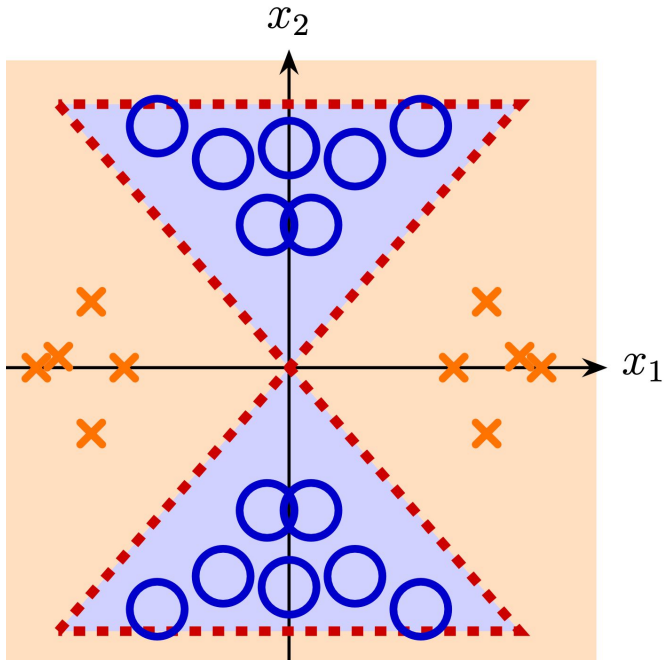


- Fill the region with many small circles
- **One-hidden-layer MLPs are universal classifiers**

[Whiteboard]

MLP for classification





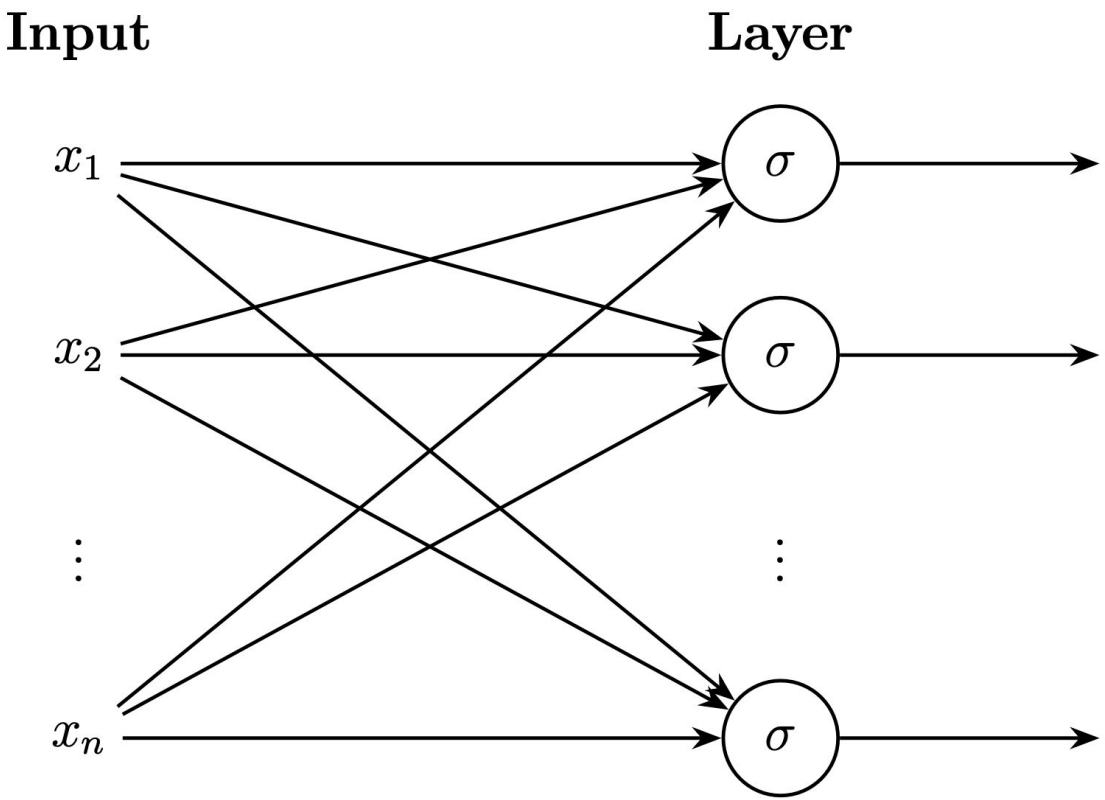
Deeper MLPs tend to require fewer neurons

[Whiteboard]

MLP for classification

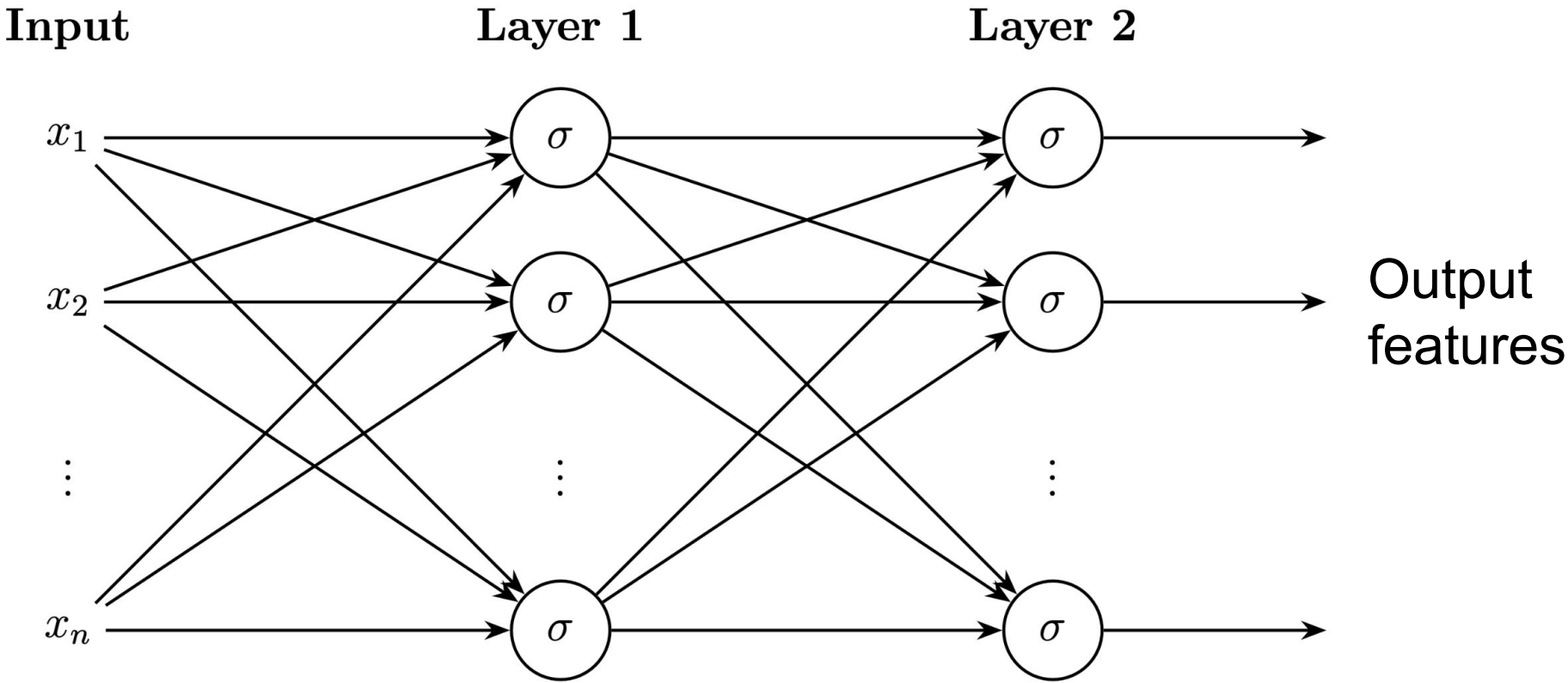
- A one-hidden-layer MLP can model any classification boundary
- MLPs are universal classifiers
- Deep MLP may require fewer neurons

A layer of multiple perceptrons

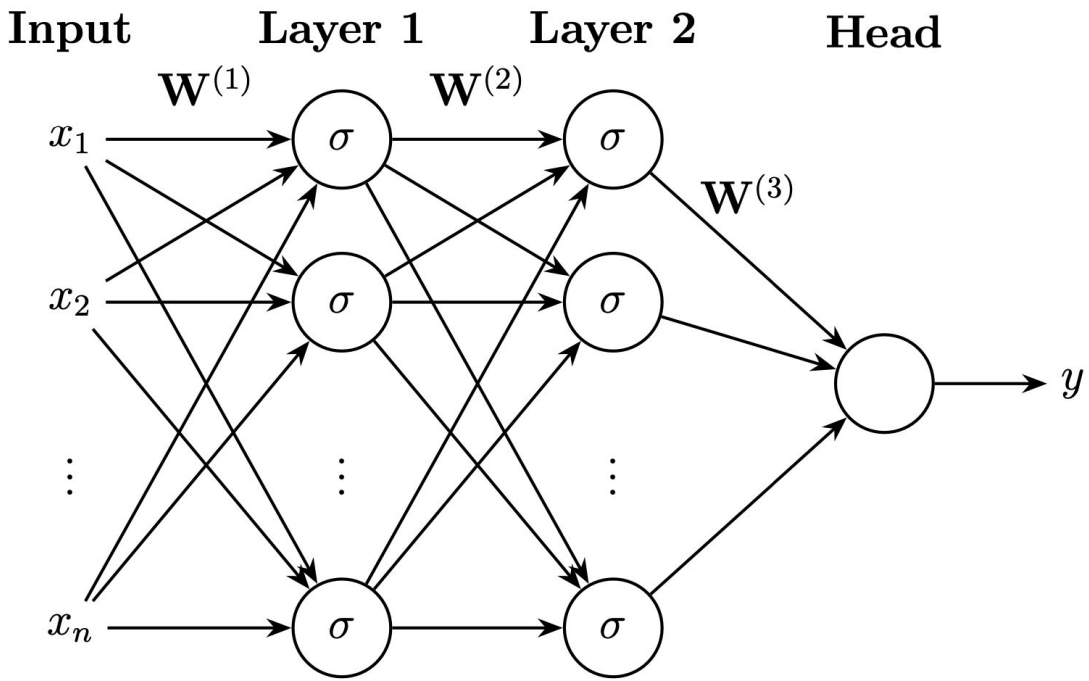


Output features

A network of multiple layers

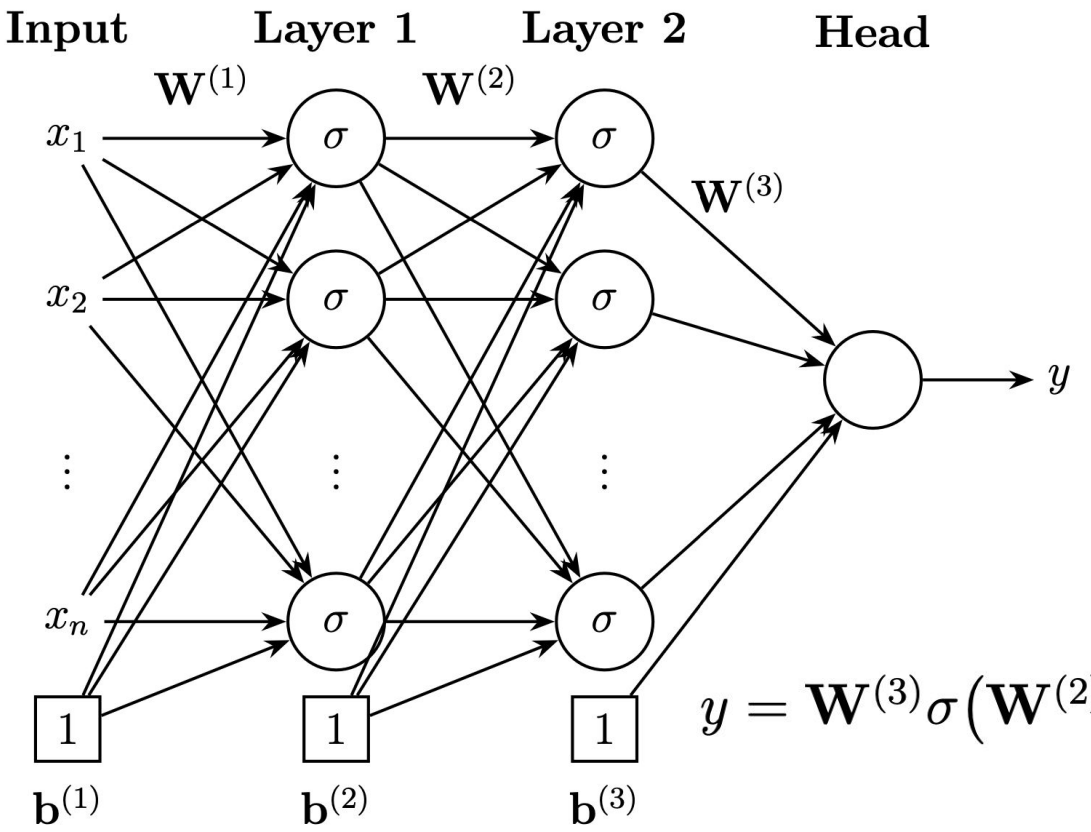


A network of multiple layers



$$y = \mathbf{W}^{(3)}\sigma\left(\mathbf{W}^{(2)}\,\sigma\left(\mathbf{W}^{(1)}\mathbf{x}\right)\right)$$

A network of multiple layers



$$y = \mathbf{W}^{(3)}\sigma\left(\mathbf{W}^{(2)}\,\sigma\left(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}\right) + \mathbf{b}^{(2)}\right) + b^{(3)}$$

Summary

- MLPs are universal function approximators and universal classifiers
- Deeper MLP are more expressive and typically require fewer neurons